

NPTEL Course on "Computer System Optimization for ML"

Lecture #42

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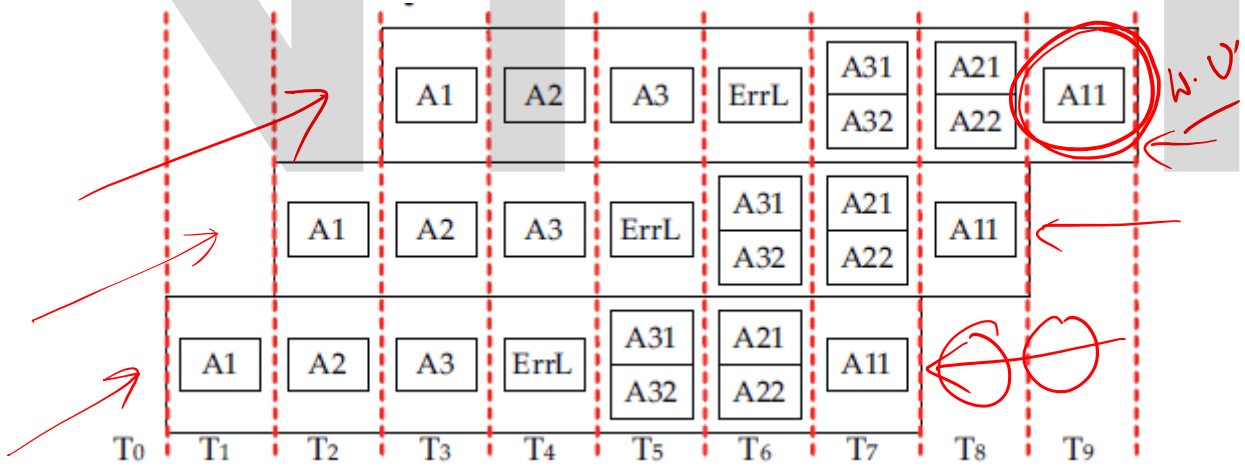
Agenda

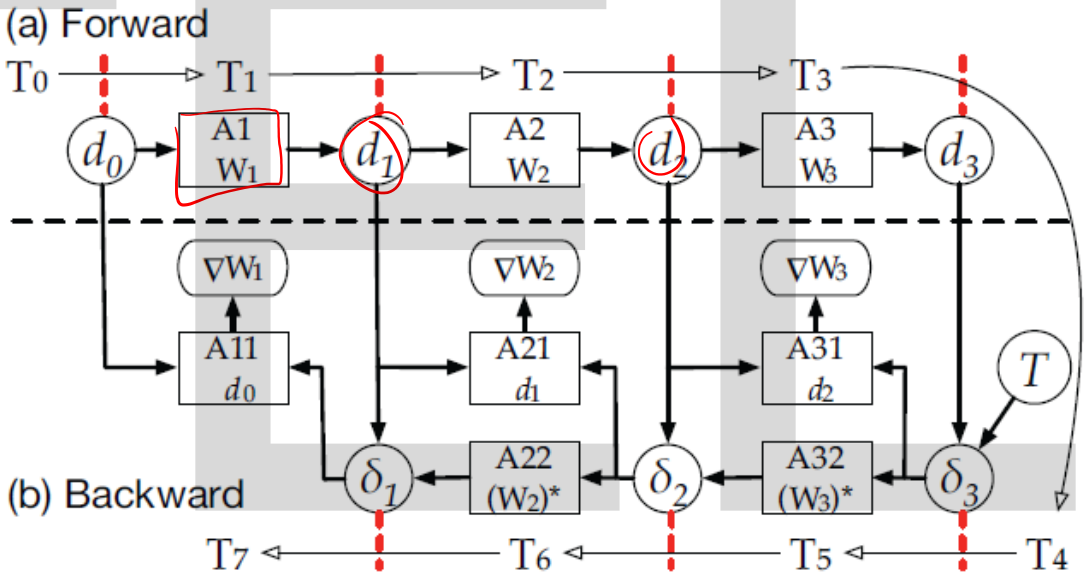
- Pipelayer (contd.)
- Atomlayer
- Simulation Platform

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Training in PipeLayer

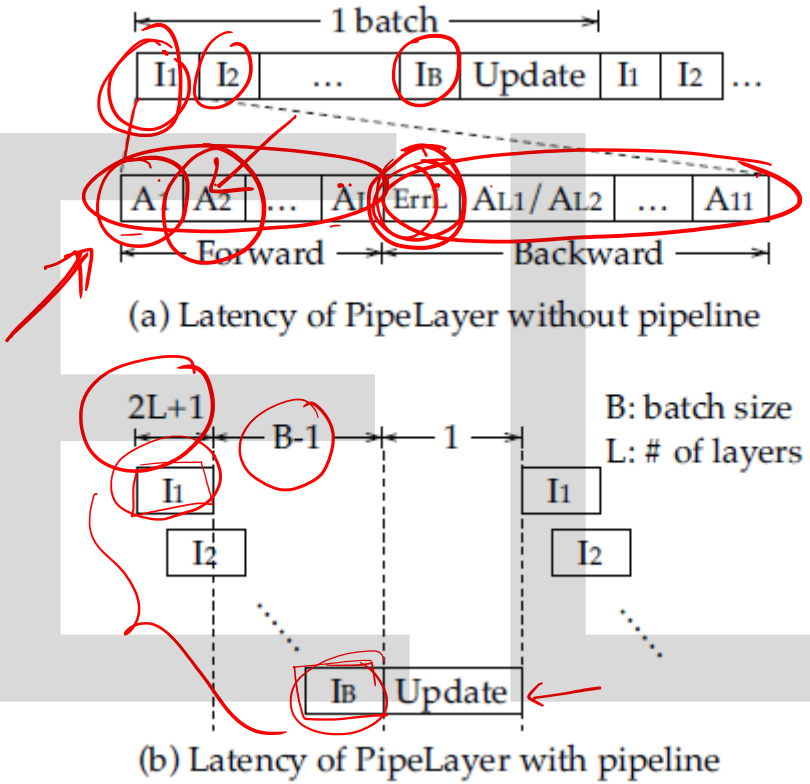
- A new input can enter the pipeline every cycle within a batch
- At the end of a batch, a new input belonging to the next batch cannot enter the pipeline until all inputs in the previous batch are processed



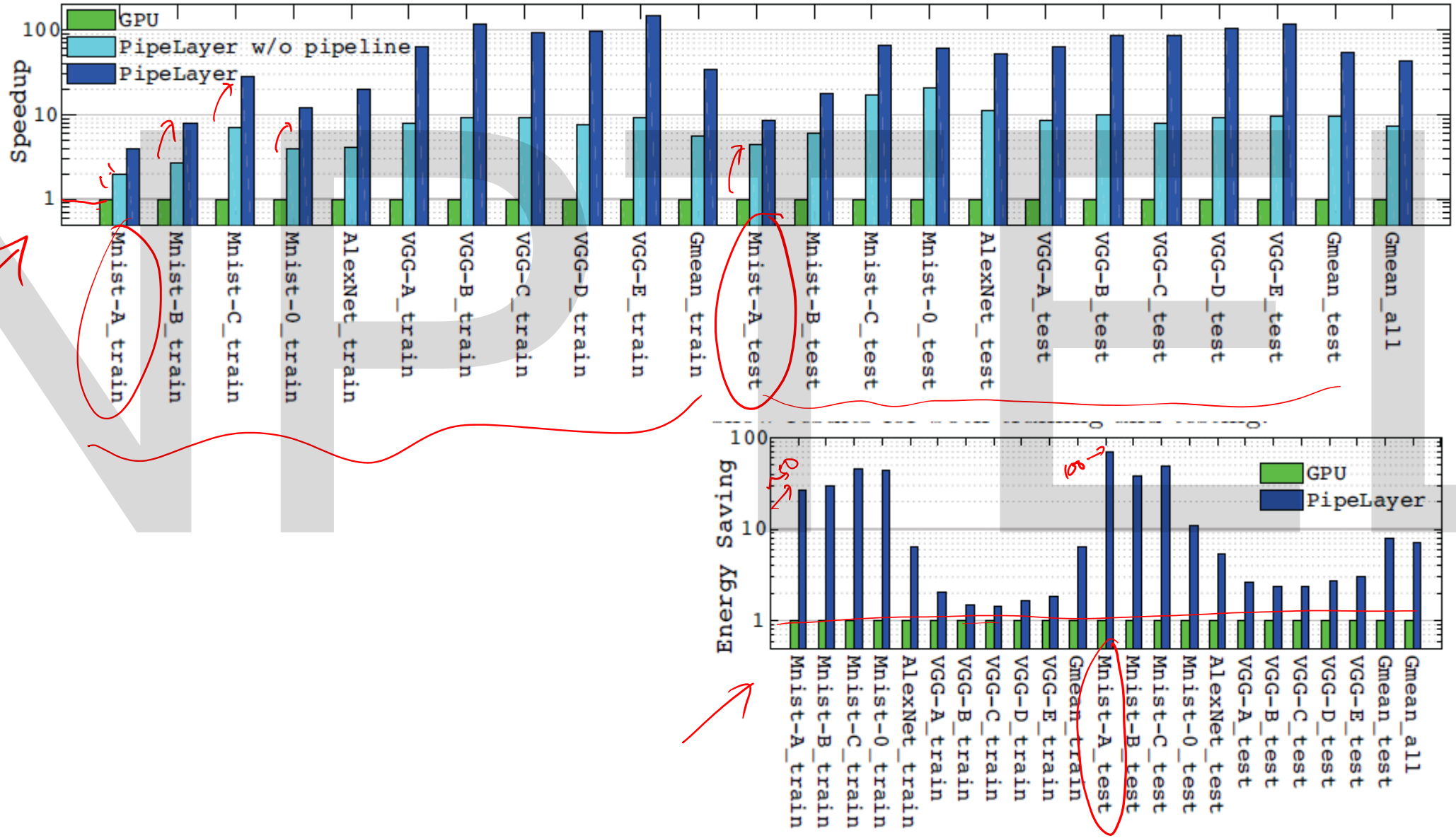


Non-pipelined vs Pipelined: Comparison

- **Assume the DNN has L layers and each layer gets executed at each (logical) cycles**
  - Forward computation: L cycles
  - Backward computation: (L+1) cycles
- **For the whole batch**
  - (2L+1)B + 1 cycles
- **For N inputs**
  - (2L+1)N + N/B cycles
- **In pipelined**
  - (N/B)(2L+B+1)



Results


GPU
PipeLayer w/o pipeline
PipeLayer
Speedup
100
10
1
Mnist-A_train
Mnist-B_train
Mnist-C_train
Mnist-0_train
AlexNet_train
VGG-A_train
VGG-B_train
VGG-C_train
VGG-D_train
VGG-E_train
Gmean_train
Mnist-A_test
Mnist-B_test
Mnist-C_test
Mnist-0_test
AlexNet_test
VGG-A_test
VGG-B_test
VGG-C_test
VGG-D_test
VGG-E_test
Gmean_test
Gmean_all
Energy Saving
GPU
PipeLayer

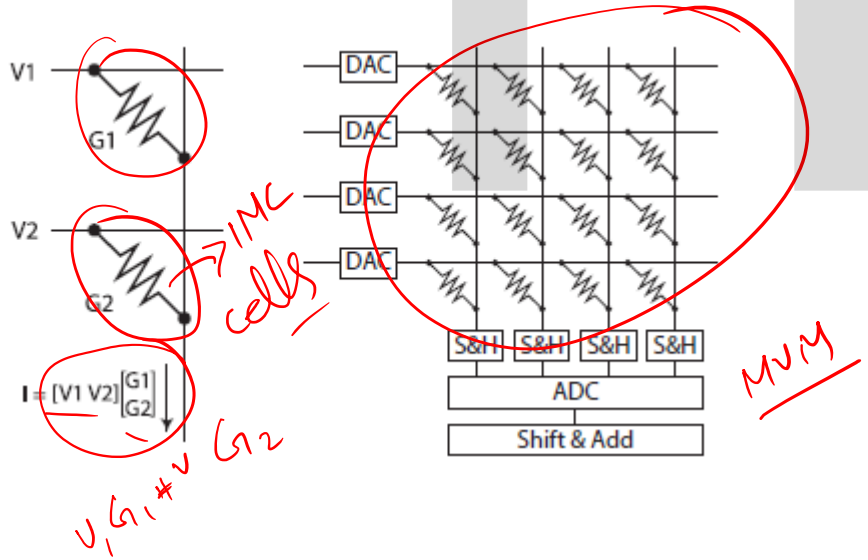
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AtomLayer: A Universal ReRAM-Based CNN Accelerator with Atomic Layer Computation

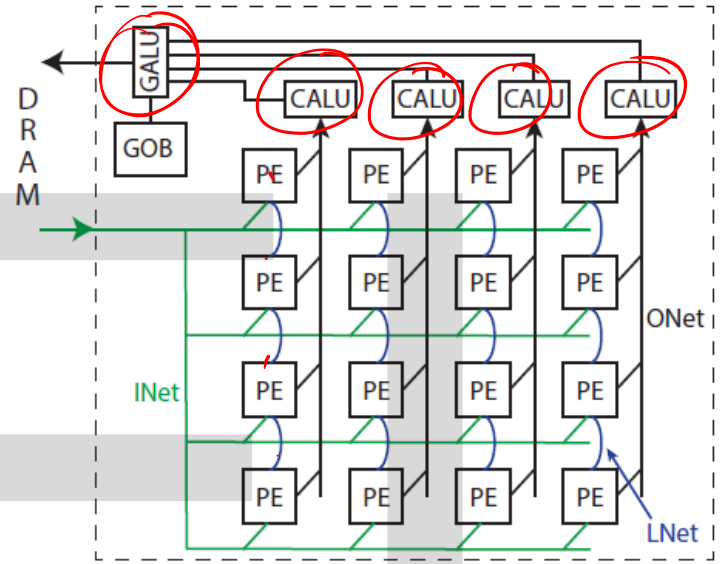
- Atomic layer computation
  - Only one layer at a time (going back to non-pipeline ☺)
  - Eliminates long latency, pipeline bubbles, large on-chip storage requirement
- Rotating crossbars structure
- Row-disjoint filter mapping and a data reuse system

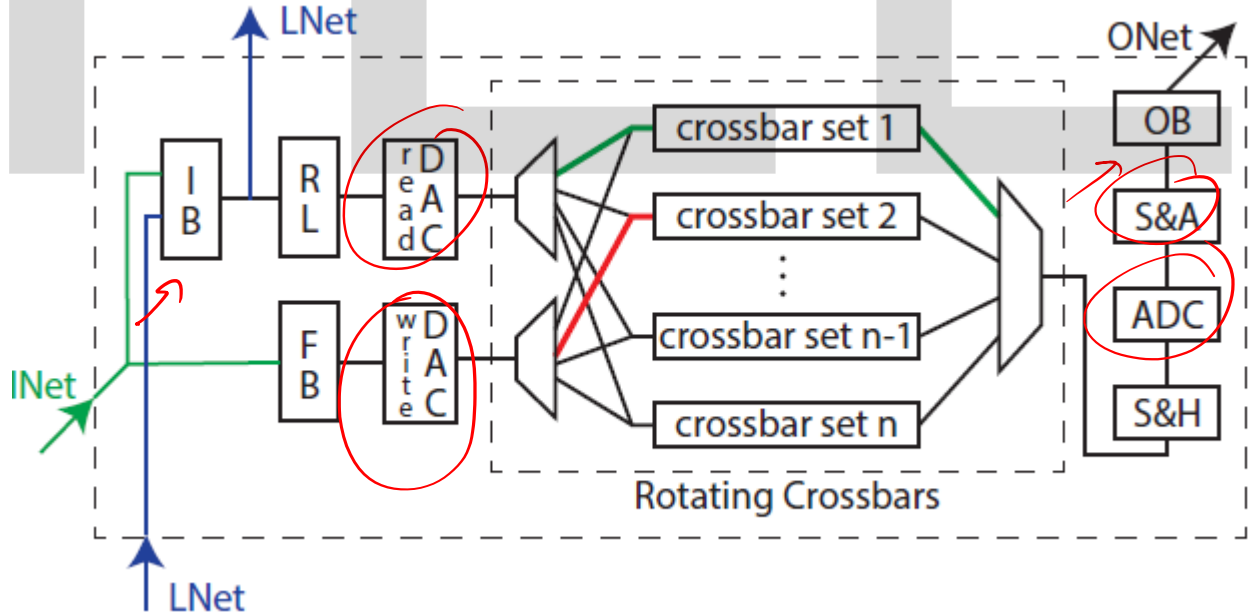


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AtomLayer Architecture

- **Processing elements (PE), on-chip network, ALU tree, global output buffer**
- **ALU tree**
  - Column ALU (CALU)
  - Global ALU (GALU)
- **PE**
  - Rotating crossbars
  - Peripheral devices
  - Four buffers

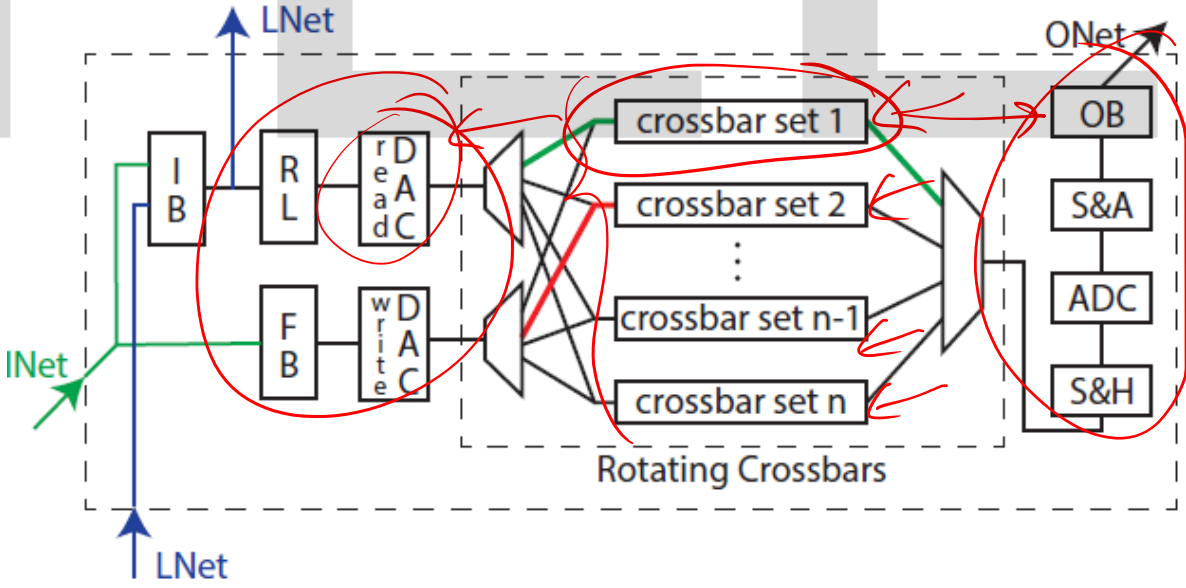




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Rotating Crossbar in AtomLayer

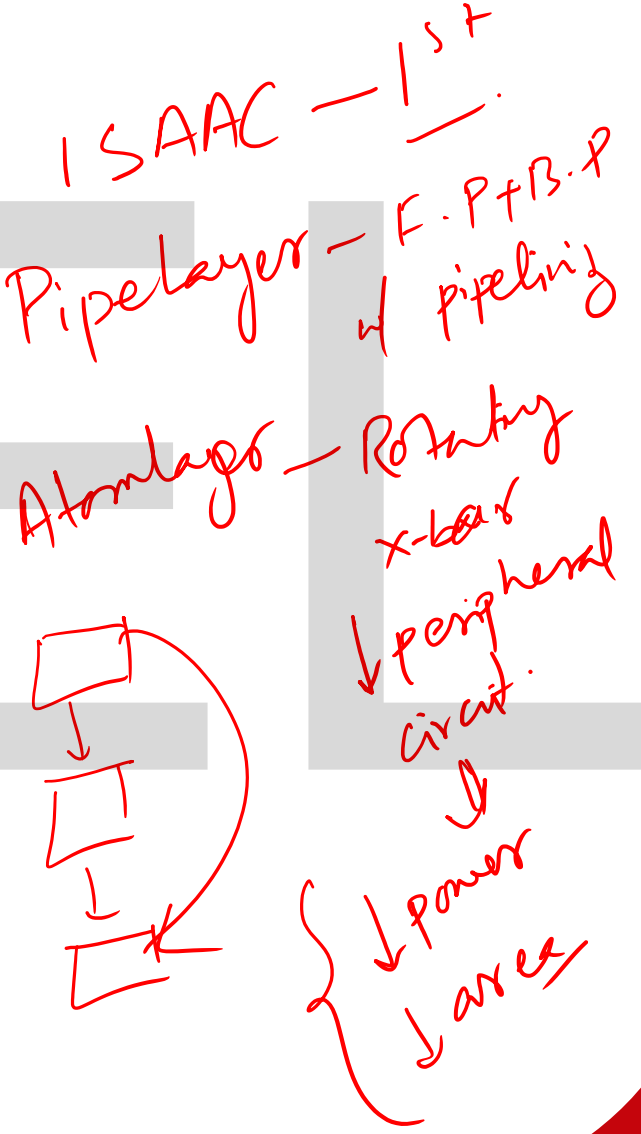
- **Key component to perform atomic layer computation**
- **Crossbars are divided into crossbar set**
  - Number of ReRAM to match the performance of the peripheral devices
- **In every cycle, once crossbar set is connected to read DACs and S&H for computation**
- **Number of sets is no less than number of NN layers**



Results

	GOPs/s/mm^2	GOPs/W	mm^2	W
160PE (w/o FC)	615.7	682.5	5.32	4.80
160PE (w/ FC)	475.6	682.5	6.89	4.80
ISAAC	478.9	627.5	85.4	65.8
PipeLayer	1485	142.9	82.6	168.6

	VGG-19 (ms)	ResNet-152 (ms)
160PE	6.92	4.01
ISAAC	8.00	43.46
PipeLayer	2.60	14.13



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- Atomlayer
- **Simulation Platform**

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)